// IBM Data Science Capstone

SpaceX Falcon9

Predictive Analysis of First Stage Landings:
A case study on reusability and aerospace efficiency.

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PREPARED BY PEDRO BONETTI

// Context

Objective

This is the final project for the IBM Data Science Professional Certificate, covering all tasks — from API handling and Exploratory Data Analysis with SQL and Python, to result prediction with ML models.

The main goal is to build an ML model to predict landing success, cutting costs through rocket reusability and gaining a competitive edge. This allows for more accurate commercial proposals and reduces the risk of multi-million dollar losses by spotting unfavorable launch conditions ahead of time.

README.md

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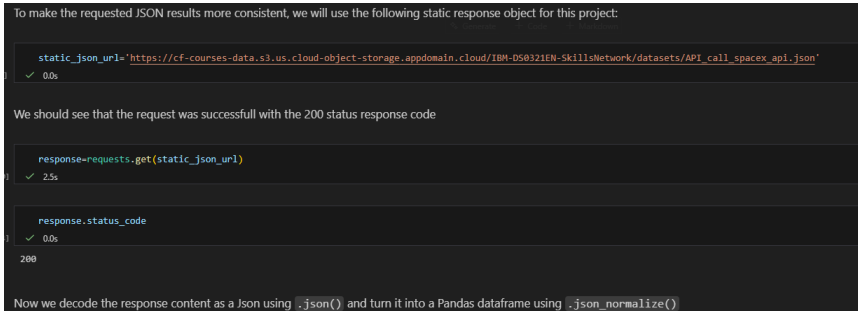
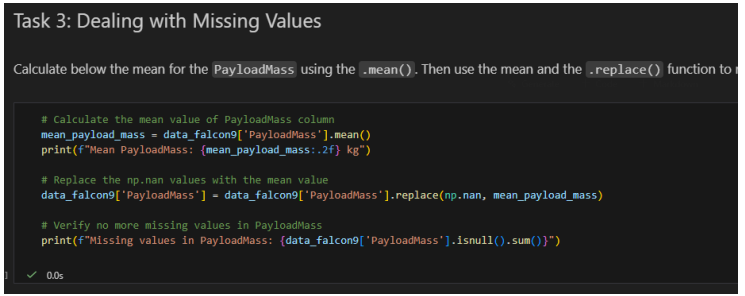
Data Collection: **REST API**





API Request

Called SpaceX's REST API (v4/launches).

JSON Decode

Decoded JSON response with pd.json_normalize().

Filtering

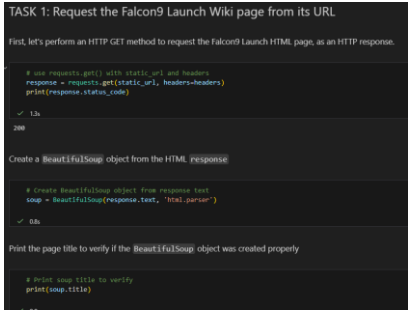
Filtered exclusively for Falcon 9 launches.

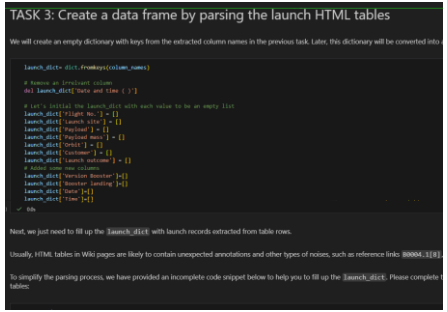
Structuring

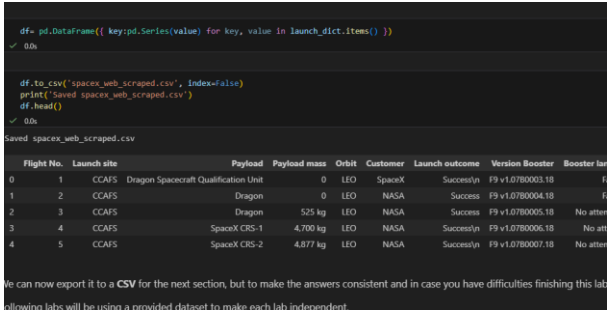
Handled nulls and formatted data types.

DataCollection-API.ipynb

Data Collection: **Web Scraping**







Wikipedia HTTP

Fetched the HTML page for Falcon 9 launches.

→

BeautifulSoup

Parsed HTML content to find elements.

→

Extraction

Mapped headers and rows to dictionaries.

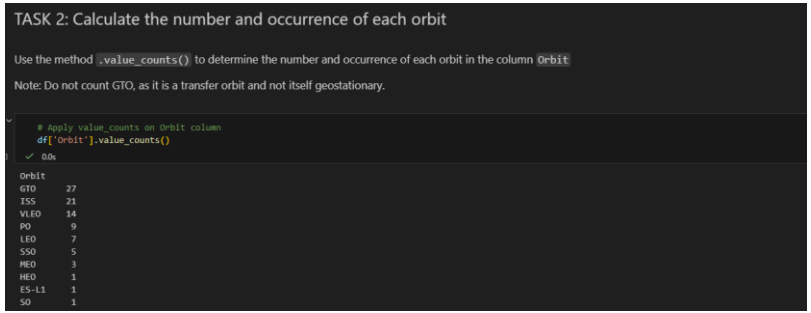
→

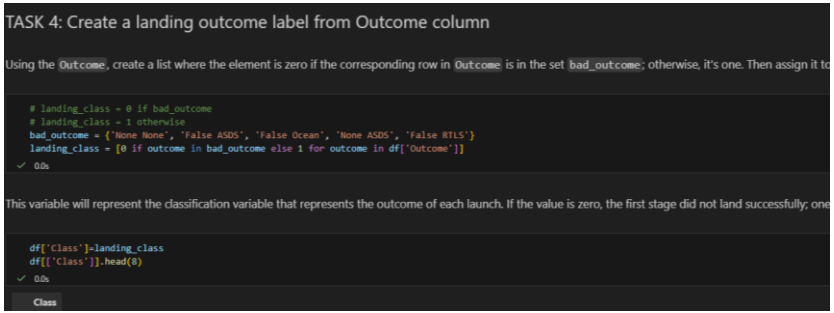
Pandas DF

Converted the dictionary into a structured DataFrame.

SpaceX-WebScraping.ipynb

Data Treatment: **Wrangling**





Missing Values

Identified missing data with .isnull().

Imputation

Calculated the mean and filled PayloadMass.

Binary Mapping

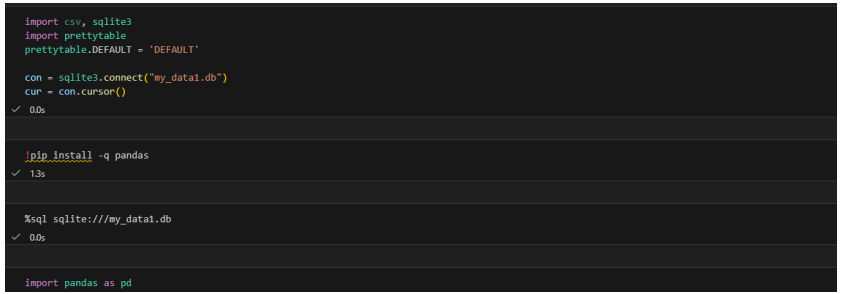
Converted outcomes into a Class variable (1=Success, 0=Failure).

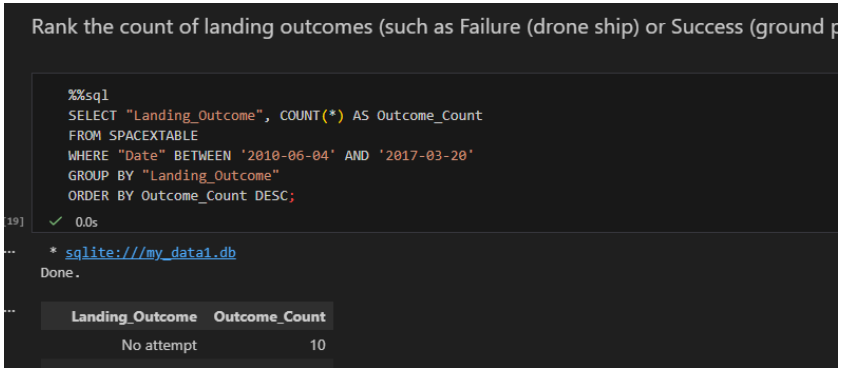
Encoding

Applied One-Hot Encoding to categorical variables.

DataWrangling.ipynb

Exploratory Analysis: **SQL**





Launch Volume: CCAFS SLC-40 handles the highest total volume of operations.

Payload Insights: Queries identified total payload mass and isolated critical weight ranges.

Historical Milestones: Pinpointed the exact first successful ground landing.

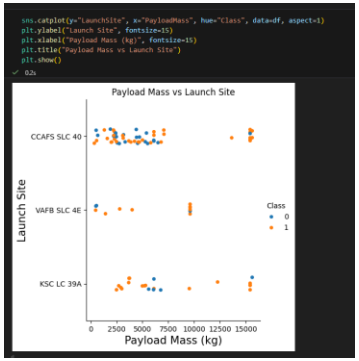
EDA-SQL.ipynb

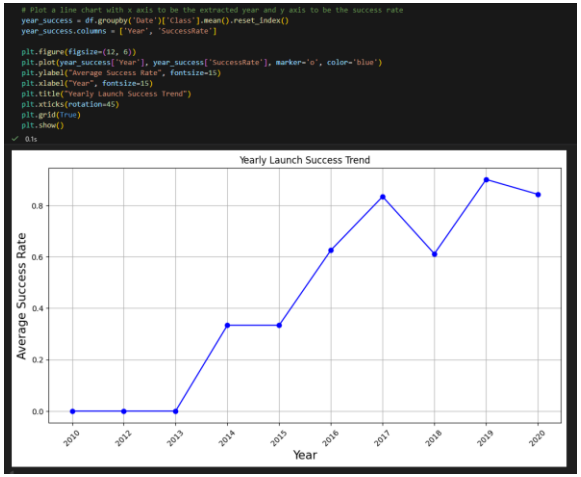
Exploratory Analysis: **Visualization**

Flight Number: Success rate improves as flight history grows (operational maturity).

Successful Orbits: ES-L1, GEO, HEO, and SSO orbits hold a 100% historical success rate.

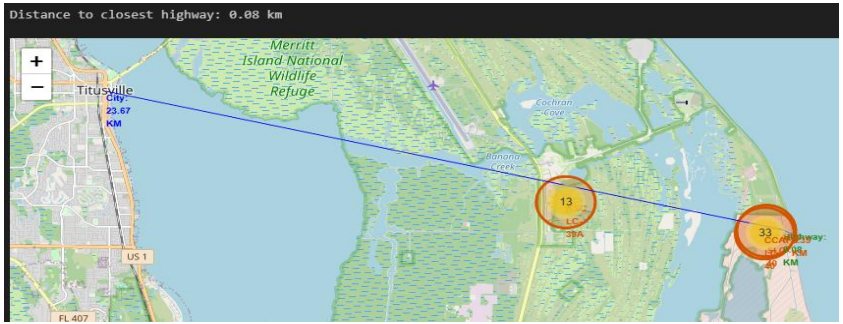
Payload Mass: Heavy payloads are mostly operated from KSC LC-39A.





EDA-DataViz.ipynb

Geospatial Analysis: **Folium Maps**



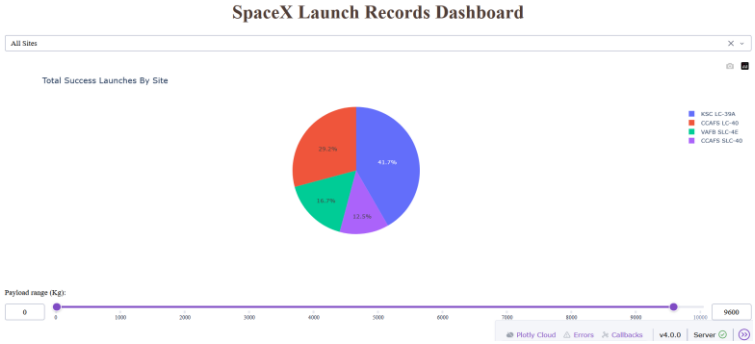
Coastal Positioning: Strategic location for safe ocean debris drop zones.

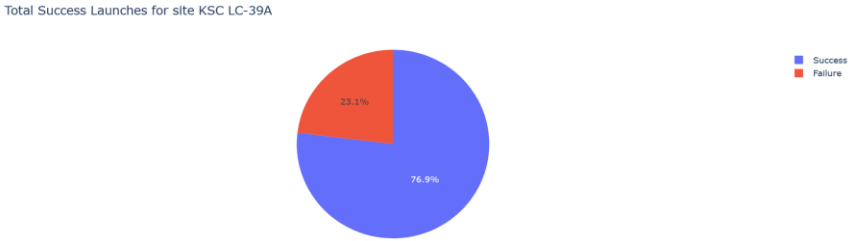
Logistics: Extremely close (< 1km) to railways and highways for booster transport.

Safety: Calculated distance from densely populated urban areas.

LauncSiteLocation.ipynb

Interactive Dashboard: **Plotly Dash**





Efficiency: KSC LC-39A has the best success-to-launch ratio.

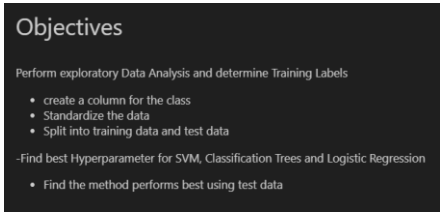
Sweet Spot: Payloads between 0–5,000 kg show the highest concentration of safe landings.

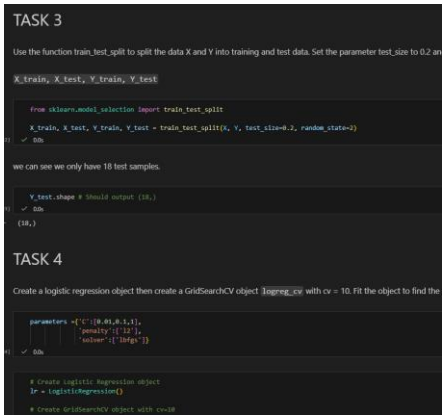
spacex_dash.py

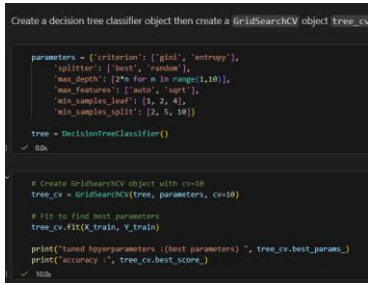
Predictive Analysis: **Classification**

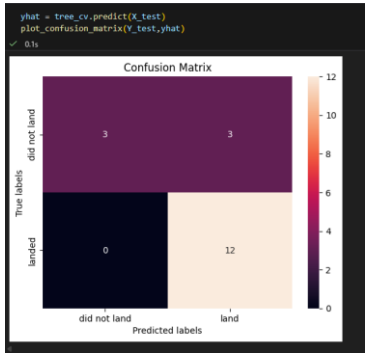
Logistic Regression	83.3%
SVM	83.3%
Decision Tree	83.3%
KNN	83.3%

Accuracy: All models tuned via GridSearchCV reached 83.3% precision on the test set.









ML-Prediction.ipynb

Project Conclusions

Strategic Site: KSC LC-39A offers the highest landing success rate.

Learning Curve: Accuracy grows with flight number, validating SpaceX's iterative engineering approach.

High Predictability: The ML models predict outcomes with 83.3% accuracy.

Business Pick: The Decision Tree is recommended for its ease of explaining results to executives.